FED SUP CLASS
5310

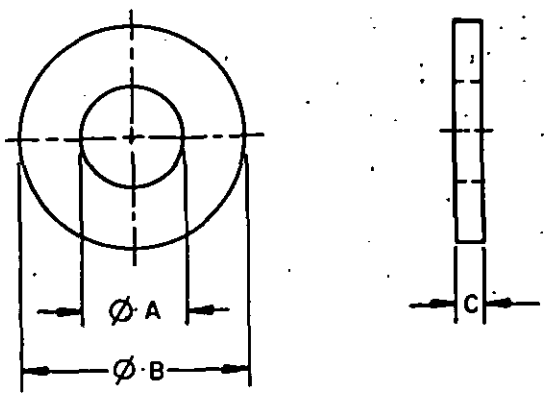



TABLE I. DASH NUMBERS AND DIMENSIONS.

| DASH NUMBERS | | Ø A | TOLERANCE | | Ø B | TOLERANCE | | C THICKNESS | | |
|---|---|---|---|---|---|---|---|---|---|---|
| CARBON STEEL | ALLOY STEEL | ID | + | − | OD | + | − | NOMINAL | MAX | MIN |
| 19 | 19A | .109 | .000 | .008 | .281 | .010 | .010 | .028 | .033 | .023 |
| 20 | 20A | .156 | .000 | .015 | .281 | .005 | .015 | .020 | .025 | .016 |
| 1 | 1A | .156 | .008 | .005 | .375 | .015 | .005 | .049 | .065 | .036 |
| 18 | 18A | .188 | .008 | .005 | .438 | .015 | .005 | .049 | .065 | .036 |
| 2 | 2A | .219 | .008 | .005 | .500 | .015 | .005 | .049 | .065 | .036 |
| 3 | 3A | .250 | .015 | .005 | .562 | .015 | .005 | .049 | .065 | .036 |
| 21 | 21A | .219 | .005 | .005 | .656 | .010 | .010 | .062 | .069 | .055 |
| 22 | 22A | .234 | .000 | .010 | .375 | .010 | .010 | .049 | .051 | .043 |
| 23 | 23A | .266 | .000 | .005 | .812 | .010 | .010 | .109 | .119 | .075 |
| 24 | 24A | .266 | .010 | .000 | 1.000 | .015 | .015 | .064 | .069 | .059 |
| 4 | 4A | .281 | .015 | .005 | .625 | .015 | .005 | .065 | .080 | .051 |
| 36 | 36A | .281 | .015 | .005 | .968 | .015 | .007 | .046 | .065 | .036 |
| 37 | 37A | .281 | .015 | .005 | 2.000 | .030 | .007 | .062 | .080 | .051 |
| 5 | 5A | .312 | .015 | .005 | .734 | .015 | .007 | .065 | .080 | .051 |
| 38 | 38A | .312 | .015 | .005 | .750 | .015 | .007 | .066 | .080 | .051 |
| 25 | 25A | .344 | .000 | .020 | 1.000 | .010 | .005 | .062 | .067 | .059 |
| 26 | 26A | .375 | .010 | .005 | .625 | .010 | .010 | .100 | .114 | .090 |
| 6 | 6A | .375 | .015 | .005 | .875 | .030 | .007 | .083 | .104 | .064 |
| 27 | 27A | .406 | .000 | .010 | 1.250 | .010 | .010 | .192 | .202 | .075 |
| 7 | 7A | .438 | .015 | .005 | 1.000 | .030 | .007 | .065 | .080 | .051 |
| 39 | 39A | .438 | .015 | .005 | 1.000 | .030 | .007 | .084 | .104 | .064 |
| 35 | 35A | .438 | .010 | .010 | 1.375 | .010 | .010 | .083 | .104 | .064 |
| 8 | 8A | .500 | .015 | .005 | 1.250 | .030 | .007 | .083 | .104 | .064 |
| 28 | 28A | .500 | .015 | .005 | .688 | .000 | .030 | .031 | .034 | .029 |
| 9 | 9A | .531 | .015 | .005 | 1.062 | .030 | .007 | .095 | .121 | .074 |
| 29 | 29A | .562 | .005 | .000 | .734 | .015 | .000 | .062 | .067 | .057 |
| 40 | 40A | .562 | .015 | .005 | 1.375 | .030 | .007 | .099 | .132 | .066 |
| 10 | 10A | .562 | .015 | .005 | 1.375 | .030 | .007 | .109 | .132 | .086 |
| 41 | 41A | .625 | .015 | .005 | 1.500 | .030 | .007 | .109 | .132 | .086 |
| 11 | 11A | .656 | .030 | .007 | 1.312 | .030 | .007 | .095 | .121 | .074 |
| 30 | 30A | .656 | .000 | .015 | 1.250 | .010 | .000 | .109 | .119 | .099 |

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APPROVED 18 OCT 82 REVISED © 16 DEC 87

USER ACTIVITIES:
ARMY-ME
NAVY-MC

REVIEWER ACTIVITIES:
ARMY-AV,EA,ER,GL
AIR FORCE-15,82
DLA-IS
NSA-NS

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AMSC N/A

| P.A. | INTERNATIONAL INTEREST | TITLE | MILITARY STANDARD |
|---|---|---|---|
| ARMY-AR<br>Other Cust<br>NAVY-SH<br>AIR FORCE-99 | | WASHER, FLAT (ROUND, STEEL, GENERAL PURPOSE) | MS51412 |
| PROCUREMENT SPECIFICATION<br>FF-W-92 | SUPERSEDES: | | PAGE 1 OF 2 |

DD FORM 672-1 (COORDINATED) PREVIOUS EDITIONS ARE OBSOLETE 5310-1592

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TABLE I. DASH NUMBERS AND DIMENSIONS -CONTINUED

| DASH NUMBERS | | Ø A | TOLERANCE | | Ø B | TOLERANCE | | C THICKNESS | | |
|---|---|---|---|---|---|---|---|---|---|---|
| CARBON STEEL | ALLOY STEEL | ID | + | − | OD | + | − | NOMINAL | MAX | MIN |
| 31 | 31A | .656 | .030 | .005 | 2.500 | .030 | .007 | .192 | .202 | .162 |
| 12 | 12A | .688 | .030 | .007 | 1.750 | .030 | .007 | .134 | .160 | .108 |
| 13 | 13A | .812 | .030 | .007 | 1.469 | .030 | .007 | .134 | .160 | .108 |
| 14 | 14A | .812 | .030 | .007 | 2.000 | .030 | .007 | .148 | .177 | .122 |
| 32 | 32A | .812 | .010 | .010 | 1.250 | .000 | .015 | .031 | .035 | .028 |
| 33 | 33A | .938 | .030 | .007 | 1.375 | .000 | .010 | .032 | .034 | .028 |
| 42 | 42A | .938 | .030 | .007 | 2.250 | .030 | .007 | .171 | .192 | .136 |
| 15 | 15A | 1.062 | .030 | .007 | 2.500 | .030 | .007 | .164 | .192 | .136 |
| 34 | 34A | 1.125 | .025 | .000 | 1.750 | .010 | .010 | .090 | .095 | .085 |
| 43 | 43A | 1.125 | .030 | .007 | 2.500 | .030 | .007 | .164 | .192 | .136 |
| 44 | 44A | 1.250 | .030 | .007 | 2.750 | .030 | .007 | .171 | .192 | .136 |
| 16 | 16A | 1.375 | .030 | .007 | 3.000 | .030 | .007 | .165 | .192 | .136 |
| 45 | 45A | 1.500 | .030 | .007 | 3.250 | .045 | .010 | .187 | .213 | .153 |
| 17 | 17A | 1.625 | .045 | .010 | 3.500 | .045 | .010 | .180 | .213 | .153 |
| 46 | 46A | 1.750 | .045 | .010 | 3.750 | .045 | .010 | .187 | .213 | .153 |
| 47 | 47A | 1.875 | .045 | .010 | 4.000 | .045 | .010 | .187 | .213 | .153 |
| 48 | 48A | 2.125 | .045 | .010 | 4.500 | .045 | .010 | .187 | .213 | .153 |
| 49 | 49A | 2.375 | .045 | .010 | 4.750 | .045 | .010 | .218 | .248 | .193 |
| 50 | 50A | 2.625 | .045 | .010 | 5.000 | .045 | .010 | .234 | .280 | .210 |

REQUIREMENTS:

1. MATERIAL: STEEL, CARBON, GRADES 1008 (UNS G10080) THRU 1020 (UNS G10200), IN ACCORDANCE WITH ASTM A108 OR ASTM A109.

   STEEL, ALLOY, GRADE 4130 (UNS G41300) IN ACCORDANCE WITH ASTM A331 OR ASTM A506.

2. PROTECTIVE COATING: ZINC COAT IN ACCORDANCE WITH ASTM B633 TYPE II, CLASS Fe/Zn 13.

3. HARDNESS: ALLOY STEEL WASHERS SHALL HAVE A HARDNESS OF 34-38HRC.

4. FLATNESS: WASHERS WITH UP TO .875 INCLUSIVE OUTSIDE DIAMETER SHALL BE FLAT WITHIN .005; WITH LARGER THAN .875 OUTSIDE DIAMETER SHALL BE FLAT WITHIN .010.

5. SURFACE ROUGHNESS: WASHERS SHALL HAVE A MAXIMUM SURFACE ROUGHNESS OF 125 MICROINCHES DETERMINED IN ACCORDANCE WITH ANSI/ASME B46.1.

6. BURRS AND SHARP EDGES: REMOVE ALL BURRS AND SHARP EDGES.

7. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER, FOLLOWED BY A DASH NUMBER FROM TABLE I.

   EXAMPLE: MS51412-20
   - 20 — DASH NUMBER
   - MS51412 — BASIC MS NUMBER

   MS51412-2 - WASHER, FLAT, CARBON STEEL, INSIDE DIAMETER .219, OUTSIDE DIAMETER .500, THICKNESS .049.

   MS51412-19A - WASHER, FLAT, ALLOY STEEL, INSIDE DIAMETER .109, OUTSIDE DIAMETER .281, THICKNESS .028.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.
4. THIS DOCUMENT WAS PREPARED FOR USE IN NON-AEROSPACE, HIGH TEMPERATURE/PETROLEUM ENVIRONMENT.

USER ACTIVITIES: ARMY-ME NAVY-MC

REVIEWER ACTIVITIES: ARMY-AV,EA,ER,GL AIR FORCE-15,82 DLA-IS NSA-NS

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| ARMY-AR<br>Order Cust:<br>NAVY-SH<br>AIR FORCE-99 | | WASHER, FLAT (ROUND, STEEL, GENERAL PURPOSE) | MS51412 |
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